Cairo University- Faculty of Engineering
Computer Engineering Department
Communications Engineering

Communications Engineering Project

AM Radio System Project

Project Overview

Design and implement an AM (Amplitude Modulation) communication system. You will explore different AM types, understand modulation index effects, and compare demodulation techniques.

System Description

Transmitter: Audio source → Bandlimit filter → AM modulator → Channel
Receiver: AM demodulator → Lowpass filter → Audio output
Modulation Types:

- **DSB-LC** (Double Sideband - Large Carrier): Conventional AM
- **DSB-SC** (Double Sideband - Suppressed Carrier): Carrier removed

Your Task

Build a working AM system using MATLAB or any other programming language and answer the questions below.

Requirements:

Audio: Use a 5-10 second mono WAV file (your choice)

1. DSB-LC vs. DSB-SC Implementation

Implement both modulation schemes: DSB-SC and DSB-LC
For DSB-LC, test three modulation indices: $\mu = 0.5, 0.8, 1.0$
For each configuration:

- Plot time-domain modulated signal
- Plot frequency-domain spectrum (magnitude)
- Measure transmitted bandwidth (99% power)

Questions:

a) Create a table comparing bandwidth for DSB-SC and all three DSB-LC cases
b) Explain: Why does DSB-LC require a carrier while DSB-SC does not?
c) Show the spectrum plots for all cases side-by-side. What is the main difference?

2. Modulation Index Effects

Using DSB-LC, test five modulation indices: $\mu = 0.3, 0.5, 0.8, 1.0, 1.2$

In this part only you may use a multitone signal consisting of the sum of two sinusoids.

For each μ:

- Plot the time-domain modulated signal
- Calculate power efficiency
- Identify overmodulation (if $\mu > 1$)

Questions:

a) Plot: Modulation index (x-axis) vs. Power efficiency (y-axis)

b) Show time-domain waveforms for $\mu = 0.5$, $\mu = 1.0$, and $\mu = 1.2$ on the same figure

c) What happens to the envelope when $\mu > 1$? Why is this a problem?

d) Explain: What is the optimal modulation index and why? Consider both power efficiency and distortion.

3. Envelope Detector Design

For DSB-LC ($\mu = 0.8$), implement an envelope detector:

Steps:

1. Rectify the signal (take absolute value or use diode model)
2. Lowpass filter to remove carrier
3. Remove DC component

Questions:

a) Plot the demodulated output and compare with the original message signal

b) Experiment with different lowpass filter cutoff frequencies. What happens if cutoff is:

- Too low (e.g., 1 kHz)?
- Too high (e.g., 50 kHz)?
- Just right (e.g., 15 kHz)?

c) Apply your envelope detector to DSB-SC instead of DSB-LC. What is the output? Why doesn't it work?

d) What is the advantage of envelope detection? What is the limitation?

4. Coherent (Synchronous) Demodulation

Implement coherent detection for both DSB-LC and DSB-SC:

Steps:

1. Multiply received signal by $2\cos(2\pi f_c t)$
2. Lowpass filter
3. Extract message

Questions:

a) Show the demodulated output for both DSB-LC and DSB-SC. Does coherent detection work for both?

b) Compare coherent detector output with envelope detector output (for DSB-LC). Which has better quality?
c) What is the main advantage of coherent detection over envelope detection?
d) What is the main disadvantage (implementation challenge)?

Deliverables:

1. An **uncompressed pdf** project report containing:
    a. Explanation of your work.
    b. All the required results and answers to questions.
    c. All the required figures. Label your figures properly.
    d. All the codes, included at the end.
2. All the code and audio files as uncompressed separate files.

Instructions:

- **Team Size:** You may work individually or in teams of 2 members maximum.
- **Academic Integrity:** Any evidence of copying (from other teams, internet, …) will result in **zero grade** for all involved teams. All work must be your own.
- **Artificial Intelligence:** The use of AI tools is **strictly prohibited** for any part of this project.
- The code given in the report should be given as text, not as screenshots.
- You should cite any references or webpages that you use.

Due date: May 8, 2026, at 11:59 PM.